

Osteoporosis

Quality standard

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This standard is based on CG76, NG5, NG56 and NG259.

This standard should be read in conjunction with QS143, QS86, QS16 and QS81.

Quality statements

Statement 1 People aged 50 and over and women who have experienced menopause, who have had a fragility fracture or use systemic glucocorticoids, have an assessment of their fragility fracture risk.

Statement 2 This statement has been removed. For more details, see [update information](#).

Statement 3 Adults prescribed drug treatment to reduce fracture risk are asked about adverse effects and adherence to treatment at each medication review.

Statement 4 Adults having long-term bisphosphonate therapy have a review of the need for continuing treatment.

Quality statement 1: Assessment of fragility fracture risk

Quality statement

People aged 50 and over and women who have experienced menopause, who have had a fragility fracture or use systemic glucocorticoids, have an assessment of their fragility fracture risk.

Rationale

Focusing assessment on people who may be at increased risk of fragility fracture, specifically those with a previous fragility fracture and those currently or frequently using systemic glucocorticoids, enables effective prevention. This reduces imminent fracture risk and morbidity, and fracture risk can be used to consider options for prevention and treatment.

Quality measures

The following measures can be used to assess the quality of care or service provision specified in the statement. They are examples of how the statement can be measured and can be adapted and used flexibly.

Process

a) Proportion of people aged 50 and over and women who have experienced menopause who have had a fragility fracture who have an assessment of their fragility fracture risk.

Numerator – the number in the denominator who have an assessment of their fragility fracture risk.

Denominator – the number of people aged 50 and over and women who have experienced menopause, who have had a fragility fracture.

Data source: Local data collection, for example, local audit of patient records. The [Royal College of Physicians' Fracture Liaison Service Database](#) collects data on people aged 50 years and over who have had a fragility fracture and were assessed by the fracture liaison service within 90 days of their fracture.

b) Proportion of people aged 50 and over and women who have experienced menopause, who use systemic glucocorticoids who have an assessment of their fragility fracture risk.

Numerator – the number in the denominator who have an assessment of their fragility fracture risk.

Denominator – the number of people aged 50 and over and women who have experienced menopause, who use systemic glucocorticoids.

Data source: Local data collection, for example, local audit of patient records.

Outcome

Incidence of fragility fractures.

Data source: Local data collection, for example, local audit of patient records. The [Royal College of Physicians' Fracture Liaison Service Database](#) collects data on people aged 50 years and over who have had a fragility fracture.

What the quality statement means for different audiences

Service providers (general practices, secondary care services and fracture liaison services) ensure that systems are in place for people aged 50 and over and women who have experienced menopause, who have had a fragility fracture or use systemic glucocorticoids to have an assessment of their fragility fracture risk.

Healthcare professionals (GPs, specialists, specialist nurses and fracture liaison practitioners) assess fracture risk, or confirm that assessment has taken place, in people aged 50 and over and women who have experienced menopause, who have had a fragility fracture or use systemic glucocorticoids to estimate their risk of fragility fracture and determine their treatment options.

Commissioners ensure that they commission services in which people aged 50 and over and women who have experienced menopause, who have had a fragility fracture or use systemic glucocorticoids have their fragility fracture risk assessed.

People aged 50 and over and women who have experienced menopause, who have had a fragility fracture in the past, or who are taking systemic steroid treatment have their risk of fragility fracture assessed. Fragility fractures happen in people with fragile bones that break easily. There are treatments available to help prevent fractures in people who are at increased risk. An assessment can help to decide if treatment will reduce the chance of having a fracture.

Source guidance

Osteoporosis: risk assessment. NICE guideline NG259 (2026), recommendation 1.1.1

Definitions of terms used in this quality statement

Fragility fracture

Fractures associated with osteoporosis, typically caused by low impact injuries that would not normally cause a fracture (such as a fall from standing height), and sometimes called low trauma fractures. Fragility fractures can occur spontaneously, in people with no history of injury. Most vertebral fragility fractures are not caused by falls, instead happening after activities involving lifting, twisting, or bending. Fragility fractures most commonly affect the spine (vertebral column), hip, shoulder (proximal humerus), and wrist (distal forearm). But fragility fractures can happen in any bone, and some fractures (for example pelvis and rib fractures) are just as strongly associated with reduced bone strength as the most common fragility fractures. Fragility fractures can occur from causes other than osteoporosis such as other bone diseases or malignancy. [NICE's guideline on osteoporosis: risk assessment, terms used in this guideline]

Use of systemic glucocorticoids

Adults with current or frequent use of systemic glucocorticoids (for example, 5 mg or more prednisolone daily or equivalent for over 3 months, or intermittent use of higher doses) unless this is a replacement dose for adrenal insufficiency (see table 2 in NICE's guideline on adrenal insufficiency). [NICE's guideline on osteoporosis: risk assessment]

recommendation 1.1.1]

Assessment of fracture risk

An assessment of fracture risk should include a full clinical risk assessment alongside a risk prediction tool to determine the 10-year risk of major osteoporotic fracture. Either FRAX or QFracture should be used within their allowed age ranges. Above the upper age limits defined by the tools, people should be considered at high risk. Measurement of bone mineral density (BMD) using a dual-energy X-ray absorptiometry (DXA) scan (with or without completing a risk prediction tool) should be offered to assess fracture risk in people aged 30 and over who have had:

- a previous hip or vertebral fragility fracture or
- a single major osteoporotic fragility fracture in the last 2 years or
- 2 or more fragility fractures.

[Adapted from [NICE's guideline on osteoporosis: risk assessment](#), recommendations 1.3.1, 1.3.2, 1.4.1 and 1.7.4]

Equality and diversity considerations

When assessing fracture risk in trans or non-binary people, an individualised risk assessment should be done that takes into account all of the following:

- sex registered at birth
- hormonal history (for example, puberty suppression, gender-affirming hormone therapy or menopause)
- current hormone treatment, if clinically relevant.

[[NICE's guideline on osteoporosis: risk assessment](#), recommendation 1.1.5]

Quality statement 2: Starting drug treatment

This statement has been removed. For more details, see [update information](#).

Quality statement 3: Adverse effects and adherence to treatment

Quality statement

Adults prescribed drug treatment to reduce fracture risk are asked about adverse effects and adherence to treatment at each medication review.

Rationale

People prescribed drugs to prevent fragility fractures sometimes stop taking them because of adverse effects. Adherence to treatment, including taking their medicine by the recommended method, is needed to ensure that fracture risk is reduced effectively. Checking how well a person is managing their treatment at each medication review means that any problems can be discussed and their treatment adjusted if needed, which will improve adherence and quality of life.

Quality measures

The following measures can be used to assess the quality of care or service provision specified in the statement. They are examples of how the statement can be measured and can be adapted and used flexibly.

Process

Proportion of medication reviews for adults prescribed drug treatment to reduce fracture risk that include a record of adverse effects and adherence to treatment.

Numerator – the number in the denominator that include a record of adverse effects and adherence to treatment.

Denominator – the number of medication reviews for adults prescribed drug treatment to reduce fracture risk.

Data source: Local data collection, for example, local audit of patient records.

Outcomes

a) Adults adhering to drug treatment to reduce fracture risk.

Data source: Local data collection, for example, local audit of patient records. The Royal College of Physicians' Fracture Liaison Service Database records adherence to drug treatment at 52 weeks.

b) Incidence of fragility fracture.

Data source: Local data collection, for example, local audit of patient records. The Royal College of Physicians' Fracture Liaison Service Database collects data on people aged 50 years and over who have had a fragility fracture.

What the quality statement means for different audiences

Service providers (general practices, secondary care services and pharmacies) ensure that systems are in place for adults prescribed drug treatment to reduce fracture risk to be asked if they have had any adverse effects and about adherence to treatment at each medication review.

Healthcare professionals (GPs, specialists, specialist nurses and pharmacists) carry out medication reviews with adults prescribed drug treatments to reduce fracture risk. At the reviews, they ask if the person has had any adverse effects and if they are taking their medicine by the recommended method and as prescribed. If any problems are raised, these should be discussed and treatment adjusted if needed, which may involve input from a specialist.

Commissioners ensure that the commission services in which adults prescribed drug treatment to reduce fracture risk are asked if they have had any adverse effects and about adherence to treatment at each medication review.

Adults taking medicine to help prevent fractures have regular medicine reviews with their doctor to check if they are having any side effects, such as heartburn or reflux, and that

they are taking the medicine correctly. The review gives the chance for any problems to be discussed and treatment can be adjusted if needed to help with side effects.

Source guidance

- [Osteoporosis – prevention of fragility fractures. NICE clinical knowledge summary \(2021, updated 2025\)](#)
- [Medicines optimisation. NICE guideline NG5 \(2015\)](#), recommendations 1.4.1 and 1.4.3
- [Medicines adherence. NICE guideline CG76 \(2009\)](#), recommendations 1.1.21 and 1.3.3

Definitions of terms used in this quality statement

Drug treatment to reduce fracture risk

Drugs that can be prescribed to prevent fragility fractures include oral bisphosphonates (alendronate, ibandronate, risedronate) and intravenous zoledronic acid, and non-bisphosphonates (raloxifene, denosumab, teriparatide, abaloparatide, romosozumab, and hormone replacement therapy). [Adapted from [National Osteoporosis Guideline Group's Clinical guideline for the prevention and treatment of osteoporosis](#), section 6]

Full details of the licensed indications for these drugs can be found in the [summary of product characteristics](#). At the time of publication (April 2017), some bisphosphonate and non-bisphosphonate drugs were off label for this use. See [NICE's information on prescribing medicines](#).

Medication review

The review should include:

- asking about adverse effects, including upper gastrointestinal adverse effects (such as dyspepsia or reflux), symptoms of atypical fracture (including new onset hip, groin, or thigh pain), and dental problems
- asking about adherence to treatment, including following the recommended method of taking the treatment

- discussing alternative treatment options if adverse effects are unacceptable or the person has difficulty adhering to treatment.

[Expert opinion and NICE's clinical knowledge summary on osteoporosis – prevention of fragility fractures]

Quality statement 4: Long-term follow-up

Quality statement

Adults having long-term bisphosphonate therapy have a review of the need for continuing treatment.

Rationale

The optimal duration of bisphosphonate therapy is unclear and there are possible adverse effects of long-term treatment. A medication review for people having long-term bisphosphonate therapy gives the opportunity to consider whether continuing treatment is the best option, or if treatment should be changed or stopped. The response to treatment may also be evaluated to help determine whether to continue treatment.

Quality measures

The following measures can be used to assess the quality of care or service provision specified in the statement. They are examples of how the statement can be measured and can be adapted and used flexibly.

Structure

a) Evidence of local arrangements to ensure that adults taking zoledronic acid for 3 years have a review of the need for continuing treatment.

Data source: Local data collection, for example, local protocols.

b) Evidence of local arrangements to ensure that adults taking alendronate, ibandronate or risedronate for 5 years have a review of the need for continuing treatment.

Data source: Local data collection, for example, local protocols.

Process

a) Proportion of adults taking zoledronic acid for 3 years who have a review of the need for continuing treatment.

Numerator – the number in the denominator who have a review of the need for continuing treatment.

Denominator – the number of adults taking zoledronic acid for 3 years.

Data source: Local data collection, for example, local audit of patient records.

b) Proportion of adults taking alendronate, ibandronate or risedronate for 5 years who have a review of the need for continuing treatment.

Numerator – the number in the denominator who have a review of the need for continuing treatment.

Denominator – the number of adults taking alendronate, ibandronate or risedronate for 5 years.

Data source: Local data collection, for example, local audit of patient records.

Outcomes

a) Patient satisfaction with long-term bisphosphonate therapy.

Data source: Local data collection, for example, patient surveys.

b) Health-related quality of life for adults having long-term bisphosphonate therapy.

Data source: Local data collection, for example, patient surveys.

What the quality statement means for different audiences

Service providers (general practices, secondary care services and pharmacies) ensure

that systems are in place for adults having long-term bisphosphonate therapy to have a review of the need for continuing treatment.

Healthcare professionals (GPs, specialists, specialist nurses and pharmacists) offer adults having long-term bisphosphonate therapy a medication review to discuss the risks and benefits of continuing treatment and assess their response to treatment, if needed.

Commissioners ensure that they commission services in which adults having long-term bisphosphonate therapy have a review of the need for continuing treatment.

Adults taking a type of medicine called a bisphosphonate over a long time to help prevent fractures have a review to discuss the risks and benefits of continuing with the treatment.

They might also have a scan to check whether their bone strength has improved to help decide whether to continue treatment.

Source guidance

- [Clinical guideline for the prevention and treatment of osteoporosis. National Osteoporosis Guideline Group \(2024\)](#), section 7, recommendations 1, 2, 4, 5 and 8
- [Multimorbidity: clinical assessment and management. NICE guideline NG56 \(2016\)](#), recommendation 1.6.16

Definitions of terms used in this quality statement

Long-term bisphosphonate therapy

Adults who have been taking alendronate, ibandronate or risedronate for 5 years or zoledronic acid for 3 years should have a review of the need for continuing treatment.

[[National Osteoporosis Guideline Group's Clinical guideline for the prevention and treatment of osteoporosis](#), section 7, recommendations 1 and 2]

Review of the need for continuing treatment

Continuation of treatment is recommended for people with any of the following risk factors:

- age over 70 years at the time that the bisphosphonate is started
- previous hip or vertebral fracture
- current treatment with oral glucocorticoids of 7.5 mg or more prednisolone/day or equivalent
- one or more fragility fractures during treatment (if treatment is not changed).

For people without risk factors, arrange a dual-energy X-ray absorptiometry (DXA) scan and consider:

- Continuing treatment if the T-score is less than -2.5, and reassessing fracture risk and bone mineral density (BMD) every 3 to 5 years.
- Stopping treatment if the T-score is greater than -2.5, and reassessing their fracture risk and BMD after 2 years.

[Adapted from [NICE's clinical knowledge summary on osteoporosis – prevention of fragility fractures](#) and [National Osteoporosis Guideline Group's Clinical guideline for the prevention and treatment of osteoporosis](#), section 7, recommendations 1 and 2]

Update information

July 2026: Changes have been made to align this quality standard with the updated [NICE guideline on osteoporosis](#). Statement 1 has been updated to reflect changes to the guidance on assessment of fragility fracture risk including populations and risk factors. Statement 2 on starting drug treatment has been removed because it now doesn't align with NICE guideline recommendations. Links, definitions and source guidance sections have also been updated throughout.

Minor changes since publication

March 2022: The data source for outcome measure a) for statement 3 has been updated.

July 2018: The [Fracture Liaison Service Database](#) has been added to the data sources for measures in statements 1 and 3.

October 2017: Information about drug licensing for bisphosphonate and non-bisphosphonate drugs has been included in the definitions section for statement 3.

June 2017: The definition of drug treatment to reduce fracture risk in quality statement 3 has been updated to remove reference to strontium ranelate. This change has been made in order to match the [National Osteoporosis Guideline Group's Clinical guideline for the prevention and treatment of osteoporosis](#).

About this quality standard

NICE quality standards describe high-priority areas for quality improvement in a defined care or service area. Each standard consists of a prioritised set of specific, concise and measurable statements. NICE quality standards draw on existing NICE or NICE-accredited guidance that provides an underpinning, comprehensive set of recommendations, and are designed to support the measurement of improvement.

Expected levels of achievement for quality measures are not specified. Quality standards are intended to drive up the quality of care, and so achievement levels of 100% should be aspired to (or 0% if the quality statement states that something should not be done). However, this may not always be appropriate in practice. Taking account of safety, shared decision-making, choice and professional judgement, desired levels of achievement should be defined locally.

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See our [webpage on quality standards advisory committees](#) for details about our standing committees. Information about the topic experts invited to join the standing members is available on the [webpage for this quality standard](#).

NICE guidance and quality standards apply in England and Wales. Decisions on how they apply in Scotland and Northern Ireland are made by the Scottish government and Northern Ireland Executive. NICE quality standards may include references to organisations or people responsible for commissioning or providing care that may be relevant only to England.

Diversity, equality and language

Equality issues were considered during development and [equality assessments for this quality standard](#) are available. Any specific issues identified during development of the quality statements are highlighted in each statement.

For all quality statements where information is given, it is important that people are provided with information that they can easily read and understand themselves, or with

support, so they can communicate effectively with health and social care services. Information should be in a format that suits their needs and preferences. It should be accessible to people who do not speak or read English, and it should be culturally appropriate and age appropriate. People should have access to an interpreter if needed. People should also have access to an advocate, if needed, as set out in [NICE's guideline on advocacy services for adults with health and social care needs](#).

For people with additional needs related to a disability, impairment or sensory loss, information should be provided as set out in [NHS England's Accessible Information Standard](#) or the equivalent standards for the devolved nations.

Commissioners and providers should aim to achieve the quality standard in their local context, in light of their duties to have due regard to the need to eliminate unlawful discrimination, advance equality of opportunity and foster good relations. Nothing in this quality standard should be interpreted in a way that would be inconsistent with compliance with those duties.

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Endorsing organisation

This quality standard has been endorsed by NHS England, as required by the Health and Social Care Act (2012)

Supporting organisations

Many organisations share NICE's commitment to quality improvement using evidence-based guidance. The following supporting organisations have recognised the benefit of the quality standard in improving care for patients, carers, service users and members of the public. They have agreed to work with NICE to ensure that those commissioning or providing services are made aware of and encouraged to use the quality standard.

- [Bone Research Society](#)
- [National Osteoporosis Society](#)
- [Royal College of Nursing \(RCN\)](#)
- [Society and College of Radiographers \(SOR\)](#)
- [British Geriatrics Society](#)